

A1: Product Ideas

Team 1

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10 Ideas

1. Pour over coffee press
2. Candle holder (adjustable light level)
3. Lamp Shade
4. Modular desk organizer (or key, wallet, etc. holder)
5. Cryptex Puzzle
6. GriGri (boat use case)
7. Compact wallet
8. Reading light that is also a bookmark
9. Car phone holder
10. Detachable anti-cootie handle for busses and trains

Top 3

1. Coffee Press
2. Cryptex Puzzle
3. GriGri

See proposals below

PROPOSAL 1: The Hand Ram Coffee Press

Description: Coffee machines can take up a lot of valuable counter space in the kitchen, especially in small homes.

The Hand Ram Coffee Press is a small portable device that allows users to easily make a quick cup of coffee in a package that can be stored in a drawer or cabinet. This design has a built-in metal filter that allows users to not have to worry about carrying separate paper filters, and reduces the number of components used in the design.

Assembly: The Hand Ram consists of the plunger, the vessel, and a gasket. The gasket is made of rubber and attaches to the plunger in a ridge that keeps it secure. The vessel consists of a lip that sits on top of a mug, and small holes that act as a filter for the coffee grounds.

Materials: Aluminum, Thermoset O-ring

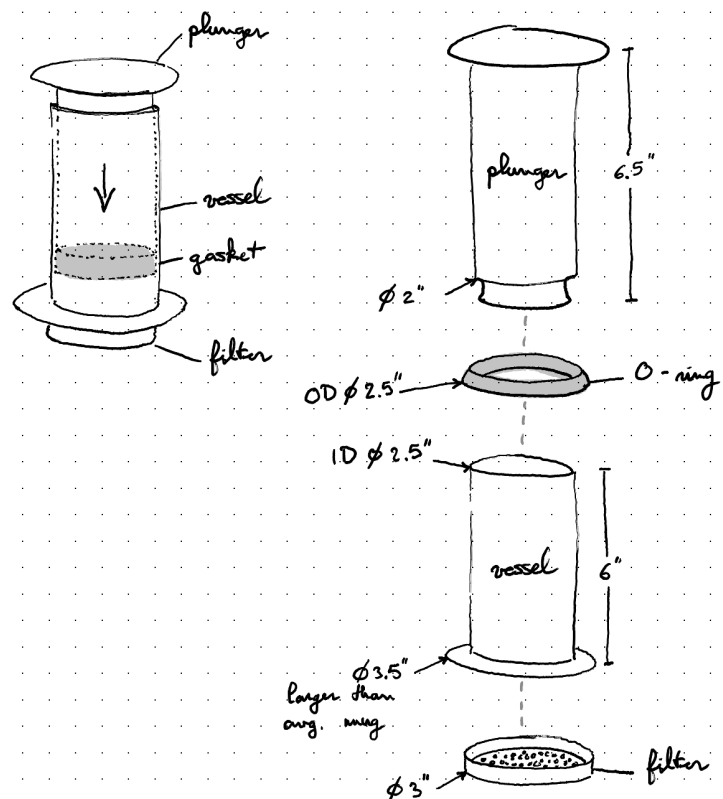
Manufacturing Process:

- Sand cast aluminum for both vessel and plunger
- CNC lathe to finish inside of vessel (may require tool to reach) and outside of plunger
- Secondary Process: Polish inside of vessel for proper seal with O-ring
- CNC mill to drill holes in vessel for filter
- Fit O-ring to plunger

Risks:

- Liquid may leak through the gasket, resulting in hot coffee/water going up along the Hand Ram and onto the user's hand
- Filter may not be possible to manufacture

Coffee Press



PROPOSAL 2: Cryptex

Description: A cryptex is a portable vault popularized by Dan Brown's novel The Da Vinci Code. With 6 unique parts (5 manufactured, 1 purchased), it creates a cylindrical compartment that is locked using rotating numbered or lettered disks.

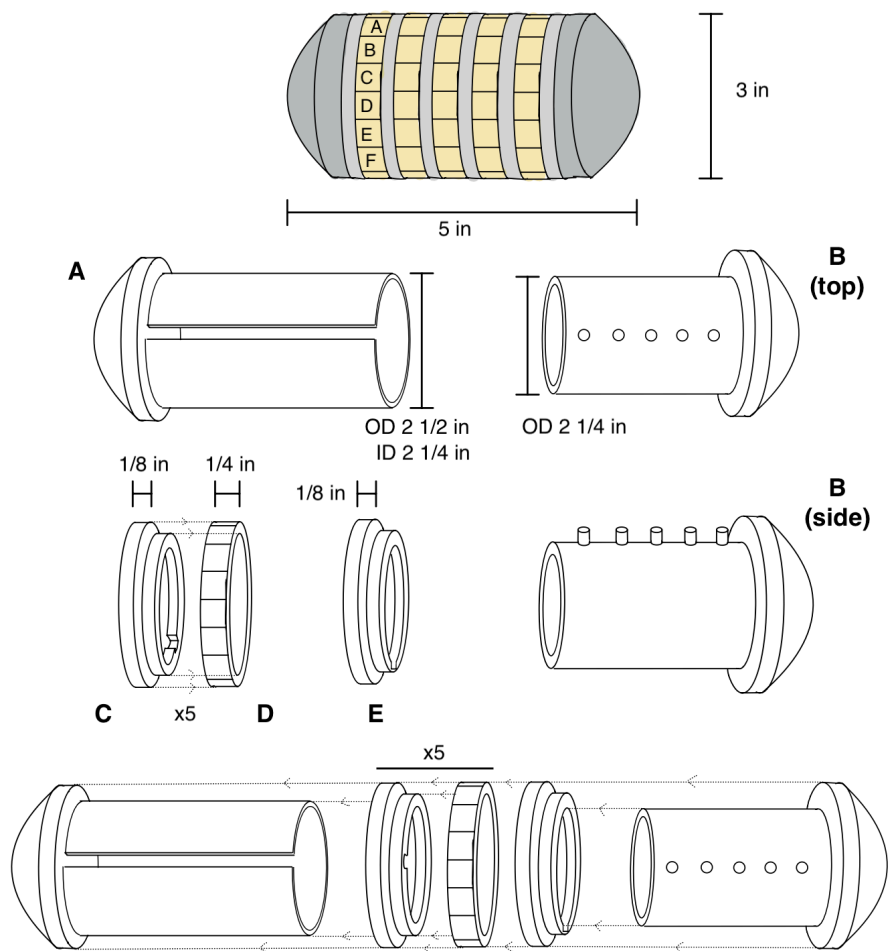
The internal locking mechanism requires tightly-fit number/letter dials and spacers. These parts will require higher tolerances. The size of the puzzle can vary, however it is usually small (~ 6 inches in length) to remain portable.

Assembly and Materials:

1. Part A: Outer Core - Aluminum
2. Part B: Inner Core - Aluminum
3. Part C: Spacer - Plastic
4. Part D: Dial - Aluminum
5. Part E: Locking Ring - plastic
6. Purchased part: set screw for Part E

Manufacturing Process:

- Cast Aluminum for parts A and B
- CNC lathe Parts A, B, and D to specifications
- CNC mill Part A to add the slit.
- CNC mill Part D to add flat sides / lettering around the outside
- Secondary Process: Polish Part D
- Injection mold Part C and E

**Risks:**

- Tolerances are a significant aspect in order to make a working lock mechanism
- Complex design and manufacturing process
- 5 unique manufactured parts, plus additional duplicate parts depending on number of dials may be too complex for the scope of this project

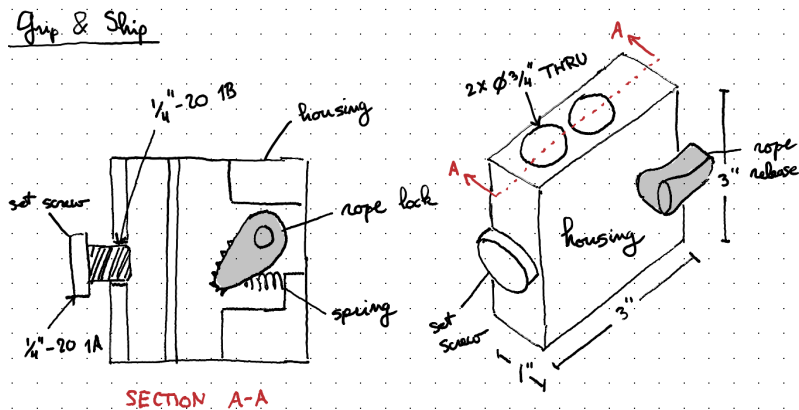
PROPOSAL 3: The Grip & Ship

Description: Many boat operators have to learn many different types of knots to tie their boats to docks. Without practice, many lose the knowledge of how to tie many of these knots.

The Grip & Ship helps operators tie their boats to pilings and cleats with ease! Rope can be permanently secured via a set screw, and then looped back into the device allowing for fast securing of boats, fenders, or any other objects. The release handle allows the operator to quickly release the rope from the Grip and Ship.

Assembly: The Grip & Ship is made up of one solid block of steel (the body) that has its holes cut out for the rope and locking mechanism

The set screw attaches to the body through a threaded hole in the body. The rope lock is held via a keyed pin that goes through the body and into the release handle. The spring maintains pressure onto the rope in the device for securing the rope.



Materials: Steel

Manufacturing Process:

- CNC mill body, add threading to hole for set screw afterwards
- CNC lathe end screw, add threads afterwards
- CNC mill rope lock and release handle
- Secondary Process: Add a plating or coating to components for salt-water corrosion resistance

Risks:

- Highly Corrosive Environment
- Spring compression strength strong enough to hold the rope, but simple to release.
- Rope Damage

